

# Practice questions

Week 7 and Week 8 · VLSI Design Flow: RTL to GDS

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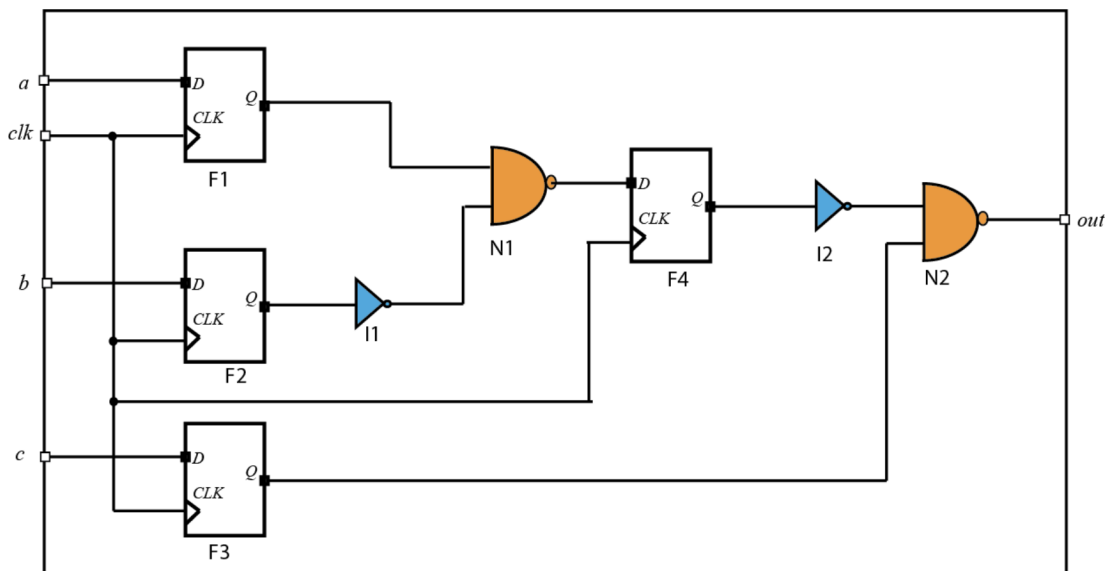
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## Week 7 : Assignment 7

1.

Consider the following synchronous circuit:



The following attributes are valid for all the flip-flops: setup time=30 ps, hold time=15 ps, and CLK-to-Q delay=40 ps. The delay of each inverter is 50 ps. The delay of each NAND gate is 80 ps. Ignore the wire delay.

Assume that the clock period is 500 ps. What is the **worst slack** for **setup** at the timing end-point F4/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 0 ps   | <input type="radio"/> 50 ps  |
| <input type="radio"/> 100 ps | <input type="radio"/> 150 ps |
| <input type="radio"/> 200 ps | <input type="radio"/> 250 ps |
| <input type="radio"/> 300 ps | <input type="radio"/> 350 ps |
| <input type="radio"/> 400 ps | <input type="radio"/> 450 ps |
| <input type="radio"/> 500 ps |                              |

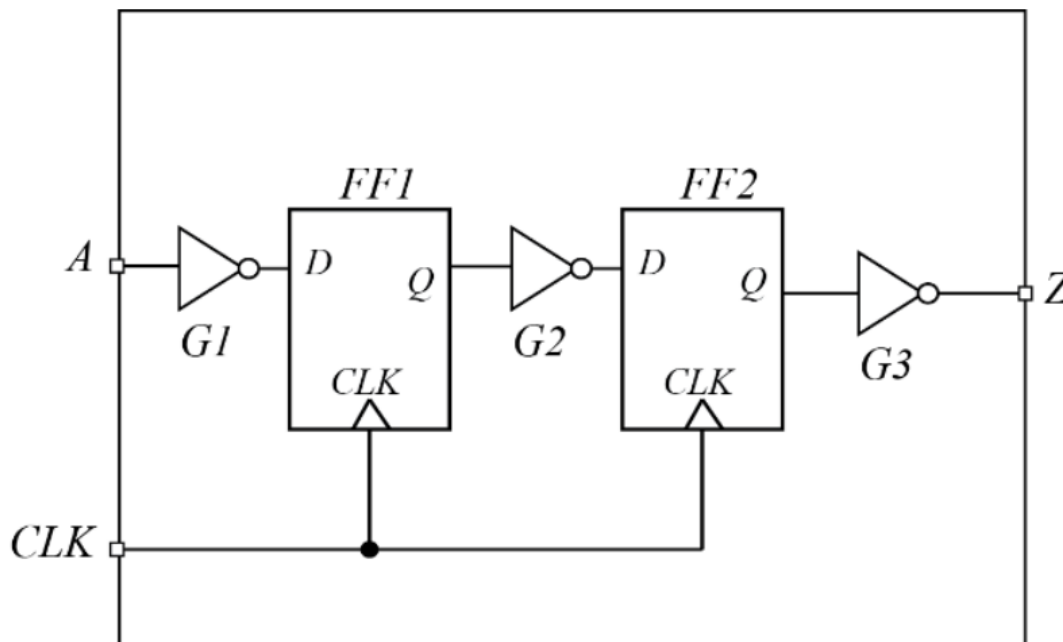
2.

For the same question above (i.e. Q 1), what is the **worst slack** for **hold** at the timing end-point F4/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 100 ps | <input type="radio"/> 105 ps |
| <input type="radio"/> 110 ps | <input type="radio"/> 115 ps |
| <input type="radio"/> 120 ps | <input type="radio"/> 125 ps |
| <input type="radio"/> 130 ps | <input type="radio"/> 135 ps |
| <input type="radio"/> 140 ps | <input type="radio"/> 145 ps |
| <input type="radio"/> 150 ps | <input type="radio"/> 155 ps |
| <input type="radio"/> 160 ps |                              |

3.

Consider the following synchronous circuit:



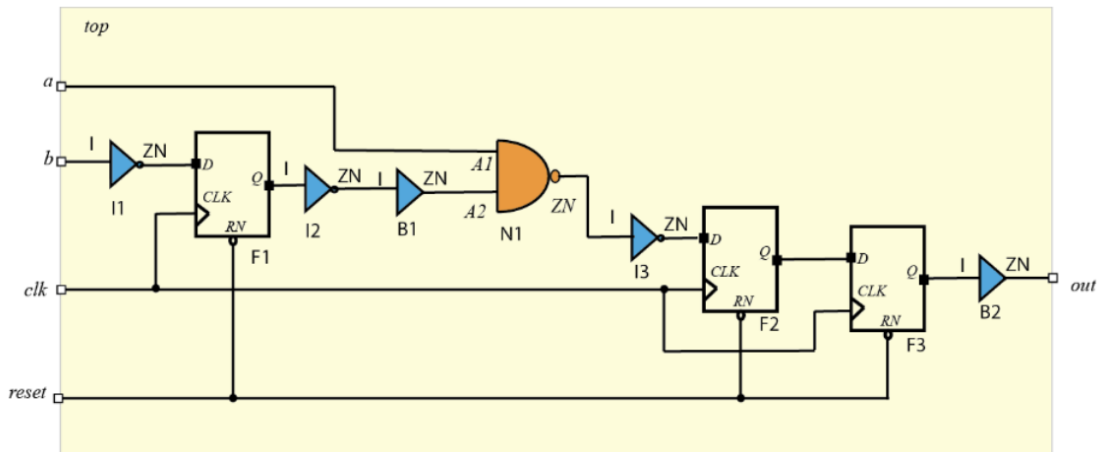
The following attributes are valid for the flip-flops FF1 and FF2: setup time=40 ps, hold time=30 ps, and CLK-to-Q delay=15 ps. The delay of each inverter is 15 ps. Ignore the wire delay.

Assume that the period of the clock is 500 ps. What is the **hold slack** at the timing end-point FF2/D?

- |                              |                             |
|------------------------------|-----------------------------|
| <input type="radio"/> -10 ps | <input type="radio"/> -5 ps |
| <input type="radio"/> 0 ps   | <input type="radio"/> 5 ps  |
| <input type="radio"/> 10 ps  | <input type="radio"/> 15 ps |
| <input type="radio"/> 20 ps  | <input type="radio"/> 25 ps |
| <input type="radio"/> 35 ps  |                             |

4.

Consider the following synchronous circuit.



The following attributes are valid for all the flip-flops: setup time=30 ps, hold time=30 ps, and CLK-to-Q delay=20 ps. The delay of each inverter and buffer is 45 ps. The delay of NAND gate is 60 ps. Ignore the wire delay.

Assume that the period of the clock is 500 ps. What is the **setup slack** at the timing end-point F3/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 0 ps   | <input type="radio"/> 50 ps  |
| <input type="radio"/> 100 ps | <input type="radio"/> 150 ps |
| <input type="radio"/> 200 ps | <input type="radio"/> 250 ps |
| <input type="radio"/> 300 ps | <input type="radio"/> 350 ps |
| <input type="radio"/> 400 ps | <input type="radio"/> 450 ps |
| <input type="radio"/> 500 ps |                              |

5.

For the same question above (i.e. Q 4), what is the **hold slack** at the timing end-point F3/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 10 ps  | <input type="radio"/> 20 ps  |
| <input type="radio"/> 30 ps  | <input type="radio"/> 0 ps   |
| <input type="radio"/> -10 ps | <input type="radio"/> -20 ps |
| <input type="radio"/> -30 ps |                              |

6.

Which of the following are typically employed in the delay calculation for a given stage during Static Timing Analysis?

A.Interconnect model

B.Driver model of the driving cell

C.Yield model for the given technology

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All
- ☐ None

7.

Which tool will you employ to compute the hold slack of a synchronous circuit?

- ☐ Icarus
- ☐ OpenSTA
- ☐ VCS
- ☐ Conformal
- ☐ Bambu

8.

The hold requirement constraint does **NOT** depend on which of the following quantities in a typical synchronous circuit?

- ☐ Clock Period
- ☐ Data Path Delay
- ☐ Delay on the clock path of the launch flip-flop
- ☐ Delay on the clock path of the capture flip-flop

9.

Consider the following statements about PBA and GBA.

A.The PBA slack can be greater than the GBA slack for some timing paths

B.The GBA slack can be greater than the PBA slack for some timing paths

C.The PBA slack can be equal to the GBA slack for some timing paths

Which of the above statements are correct?

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All
- ☐ None

**10.**

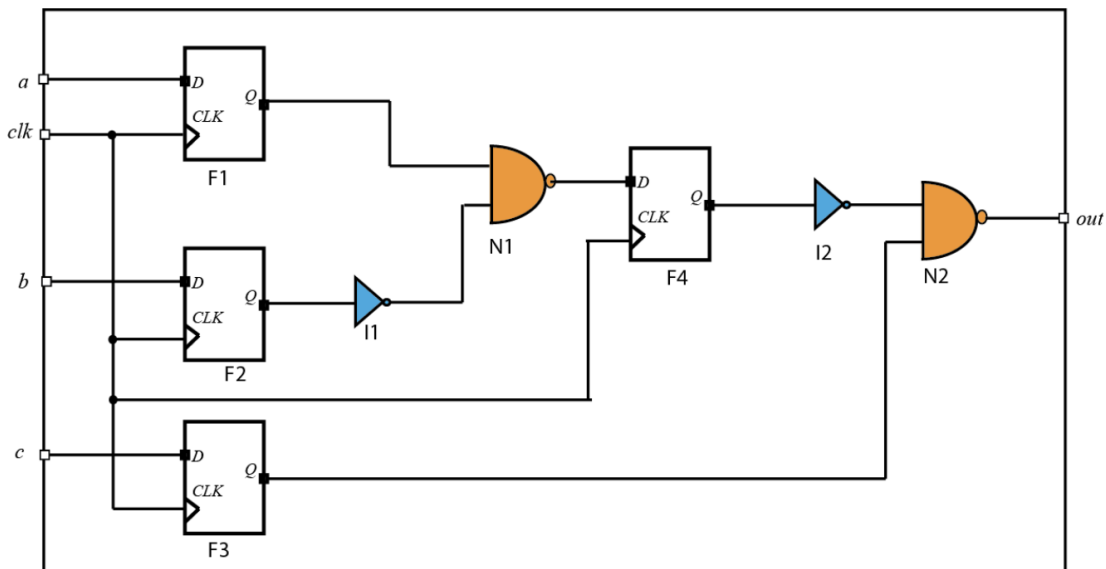
Which of the following combinations are considered when creating scenarios for multi-mode multi-corner (MMMC) analysis?

- PVT corners for libraries, RC corners for parasitics, Mode-specific SDC files for constraints
- Low VT and High VT libraries, Floorplan corners for placement, Mode-specific SDC files for constraints
- PVT corners for libraries, Floorplan corners for placement, Timing Exceptions for Constraints
- PVT corners for libraries, Floorplan corners for placement, Mode-specific SDC files for constraints

## Week 8 : Assignment 8

1.

Consider the following synchronous circuit:



The following attributes are valid for all the flip-flops: setup time=30 ps, hold time=20 ps, and CLK-to-Q delay=40 ps. The delay of each inverter is 40 ps. The delay of each NAND gate is 80 ps. Ignore the wire delay.

Assume that we have defined the following constraints in the SDC file (all time units are in picoseconds):

```
create_clock -name CLK -period 500 [get_ports clk]
set_input_delay -clock [get_clocks CLK] 150 [get_ports a]
set_input_delay -clock [get_clocks CLK] 50 [get_ports b]
set_input_delay -clock [get_clocks CLK] 200 [get_ports c]
set_output_delay -clock [get_clocks CLK] 100 [get_ports out]
```

What is the **setup slack** at the timing end-point F1/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 200 ps | <input type="radio"/> 220 ps |
| <input type="radio"/> 240 ps | <input type="radio"/> 260 ps |
| <input type="radio"/> 280 ps | <input type="radio"/> 300 ps |
| <input type="radio"/> 320 ps | <input type="radio"/> 340 ps |
| <input type="radio"/> 360 ps | <input type="radio"/> 400 ps |

2.

For the question above (Q-1), what is the **hold slack** at the timing end-point F1/D?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 100 ps | <input type="radio"/> 120 ps |
| <input type="radio"/> 130 ps | <input type="radio"/> 140 ps |
| <input type="radio"/> 150ps  | <input type="radio"/> 160 ps |
| <input type="radio"/> 170 ps | <input type="radio"/> 180 ps |
| <input type="radio"/> 190 ps | <input type="radio"/> 200 ps |

3.

For the question above (Q-1), what is the **worst slack** for setup at the timing end-point port out?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 200 ps | <input type="radio"/> 220 ps |
| <input type="radio"/> 240 ps | <input type="radio"/> 260 ps |
| <input type="radio"/> 280 ps | <input type="radio"/> 300 ps |
| <input type="radio"/> 320 ps | <input type="radio"/> 340 ps |
| <input type="radio"/> 360 ps | <input type="radio"/> 400 ps |

4.

For the question above (Q-1), what is the **worst slack** for hold at the timing end-point port out?

- |                              |                              |
|------------------------------|------------------------------|
| <input type="radio"/> 200 ps | <input type="radio"/> 220 ps |
| <input type="radio"/> 240 ps | <input type="radio"/> 260 ps |
| <input type="radio"/> 280 ps | <input type="radio"/> 300 ps |
| <input type="radio"/> 320 ps | <input type="radio"/> 340 ps |
| <input type="radio"/> 360 ps | <input type="radio"/> 400 ps |

5.

Which of the following is the SDC command that instructs an STA tool to perform a hold check at the output port?

- |                                            |                                         |
|--------------------------------------------|-----------------------------------------|
| <input type="radio"/> set_output_delay     | <input type="radio"/> set_hold_check    |
| <input type="radio"/> set_both_check       | <input type="radio"/> set_clock_latency |
| <input type="radio"/> set_clock_transition | <input type="radio"/> set_false_path    |
| <input type="radio"/> set_case_analysis    |                                         |

6.

Which of the following pieces of information is typically needed as input by a technology mapping tool?

A.Unmapped netlist

B.Technology library

C.Testbench for RTL simulation

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All
- ☐ None

7.

Which of the following optimization techniques is typically **NOT** targeted for improving timing:

- ☐ Restructuring
- ☐ Retiming
- ☐ Resizing
- ☐ Power Gating
- ☐ Fanout optimization

8.

Which of the following optimization techniques involves changing logic circuit elements across a flip-flop:

- ☐ Restructuring
- ☐ Retiming
- ☐ Resizing
- ☐ Power Gating
- ☐ Fanout optimization

9.

Which of the following is/are valid SDC commands:

A.create\_my\_clock

B.generate\_clock

C.create\_generated\_clock

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only B, C
- ☐ only A, C
- ☐ all A, B, C
- ☐ none

## 10.

Consider the following timing report generated by OpenSTA. The unit of time is ps.

| Startpoint: a (input port clocked by CLK)  |         |                               |
|--------------------------------------------|---------|-------------------------------|
| Endpoint: out (output port clocked by CLK) |         |                               |
| Path Group: CLK                            |         |                               |
| Path Type: max                             |         |                               |
| Delay                                      | Time    | Description                   |
| -----                                      |         |                               |
| 0.00                                       | 0.00    | clock CLK (rise edge)         |
| 0.00                                       | 0.00    | clock network delay (ideal)   |
| 5.00                                       | 5.00    | v input external delay        |
| 0.00                                       | 5.00    | v a (in)                      |
| 80.00                                      | 85.00   | ^ I1/ZN (INV)                 |
| 0.00                                       | 85.00   | ^ out (out)                   |
|                                            | 85.00   | data arrival time             |
|                                            |         |                               |
| 1000.00                                    | 1000.00 | clock CLK (rise edge)         |
| 0.00                                       | 1000.00 | clock network delay (ideal)   |
| 0.00                                       | 1000.00 | clock reconvergence pessimism |
| -5.00                                      | 995.00  | output external delay         |
|                                            | 995.00  | data required time            |
| -----                                      |         |                               |
|                                            | 995.00  | data required time            |
|                                            | -85.00  | data arrival time             |
| -----                                      |         |                               |
|                                            | 910.00  | slack (MET)                   |

Which of the following statements are correct?

A.The timing report shows the details of a path starting from port a and ending on port out

B.The external delay at the input port is 910 ps.

C.The report shows a hold timing violation with slack of -85 ps.

- |                                   |                                 |
|-----------------------------------|---------------------------------|
| <input type="radio"/> only A      | <input type="radio"/> only B    |
| <input type="radio"/> only C      | <input type="radio"/> only A, B |
| <input type="radio"/> only B, C   | <input type="radio"/> only A, C |
| <input type="radio"/> all A, B, C | <input type="radio"/> none      |

Source: NPTEL Week 7 : Assignment 7.

Source: NPTEL Week 8 : Assignment 8.

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